Ch8 Submanifolds, Diffeomorphisms, and the Inverse Function Theorem

Immersions, Submersions, Embeddings & Regular Submanifolds

LXJ

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Immersions, Submersions and Embeddings

Maps between manifolds

Let $\Psi : M \to N$ be a C^∞ map. Recall that the differential $\Psi_{*,m} : T_m M \to T_{\Psi(m)} N$ is a linear map.

Definition

Ψ is called

1. an immersion (浸入) if $\Psi_{*,m}$ is non-singular for all $m \in M$, i.e. $\ker \Psi_{*,m} = \{0\} \subset T_m M$;
2. a submersion (淹没) if $\Psi_{*,m}$ is a surjective map for all $m \in M$;
3. an embedding (嵌入) if Ψ is an immersion and also one-to-one, and $\Psi : M \to \Psi(M) \subset N$ is a homeomorphism with $\Psi(M)$ carrying the relative topology;
4. a diffeomorphism if Ψ is one-to-one, onto, and Ψ^{-1} is also C^∞.

Submanifolds

Definition

Let M, N be C^∞-manifolds.

1. If M is a subset of N, and the inclusion map $\iota : M \to N$ is an immersion, then M is called an *immersed submanifold* of N.
2. If the inclusion map $\iota : M \to N$ is an embedding, then M is called a *regular submanifold* of N.

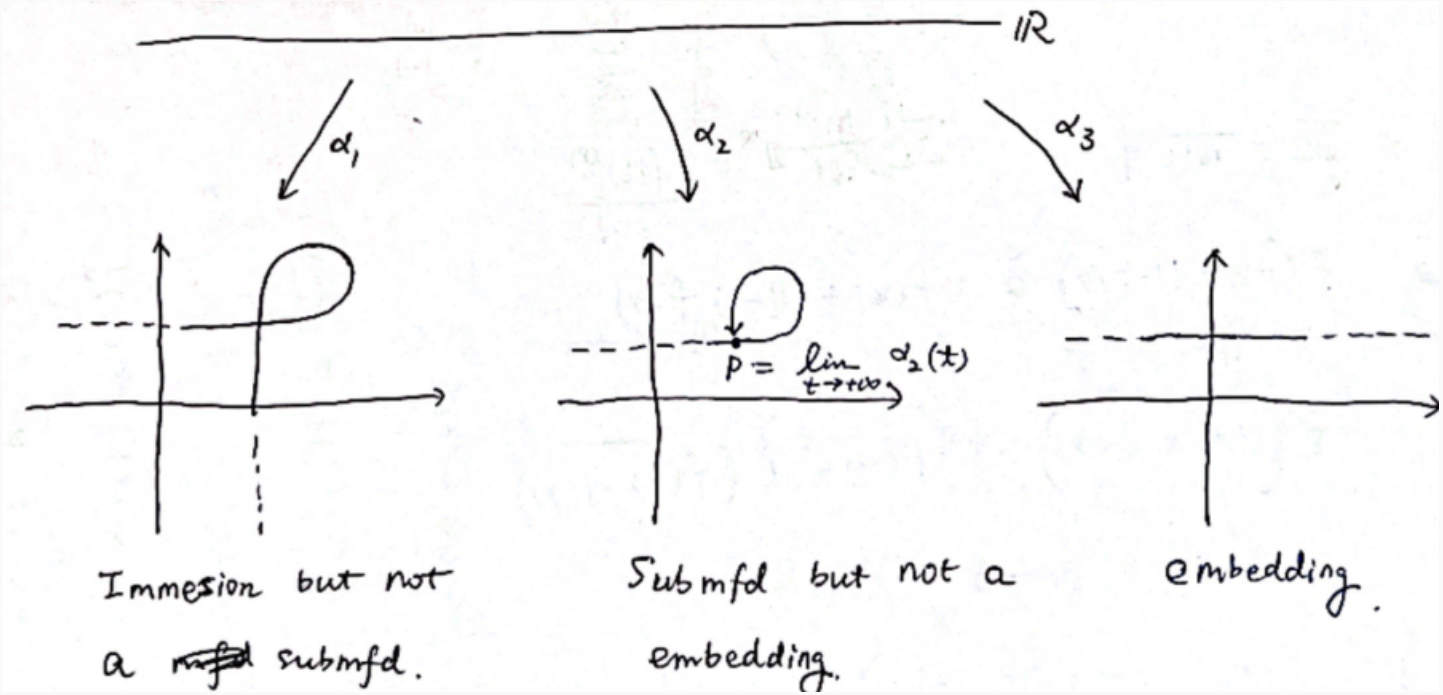
Submanifolds

Lemma

Let M and N be C^∞-map. Let $f : M \to N$ be a C^∞-map.

1. *If f is an injective immersion, considering $f(M)$ as a manifold with the topology inherited from M, then $f(M)$ is an immersed submanifold of N;*
2. *If f is an embedding ($f(M)$ has a topology inherited from N), then $f(M)$ is a regular manifold of N.*

Immersions of the real line into the plane



- α_1: immersion but not a regular submanifold;
- α_2: image is a submanifold but α_2 is not an embedding ($p = \lim_{t \to +\infty} \alpha_2(t)$);
- α_3: embedding.

Remarks on diffeomorphisms

Remark

Suppose M and N are diffeomorphic. Then N and M are diffeomorphic. It is clear that being diffeomorphic is an equivalence relation.

- **Milnor** proved that S^7 has a differentiable structure which is *not* diffeomorphic to the usual one (exotic spheres).
- There are also locally Euclidean spaces with *no* differentiable structures at all.

Examples

Examples (1–3)

Eg 1

The inclusion $\iota : S^1 \longrightarrow \mathbb{R}^2$ is an embedding.

Eg 2

$f : \mathbb{R} \to \mathbb{R}^2$, $t \longmapsto (\cos t, \sin t)$ is an immersion but not an embedding (not one-to-one).

Eg 3 (Injective immersion)

$f : \mathbb{R} \to \mathbb{R}^2$,

$$t \longmapsto \left(\frac{t^3 + t}{t^4 + 1}, \frac{t^3 - t}{t^4 + 1} \right) \qquad (t \in \mathbb{R}).$$

f is one-to-one and f_* is non-singular, but f is not a homeomorphism between $\mathbb{R}$ and $f(\mathbb{R})$ because $f(\mathbb{R})$ is compact. Hence f is not an embedding.

Examples (4–6)

Eg 4 (Embedding of the torus)

$f : T^2 = S^1 \times S^1 \to \mathbb{R}^3$,

$$(e^{i\theta}, e^{i\varphi}) \longmapsto \big((a + b\cos\varphi)\cos\theta,\ (a + b\cos\varphi)\sin\theta,\ b\sin\varphi\big).$$

f is a smooth embedding; $f(T^2)$ is a torus in $\mathbb{R}^3$.

Eg 5 (Standard embedding, $m \leq n$)

$f : \mathbb{R}^m \to \mathbb{R}^n, \quad (x^1, \ldots, x^m) \longmapsto (x^1, \ldots, x^m, 0, \ldots, 0).$

Eg 6 (Standard submersion, $m > n$)

$f : \mathbb{R}^m \to \mathbb{R}^n, \quad (x^1, \ldots, x^m) \longmapsto (x^1, \ldots, x^n).$

Normal Form for an Immersion

The local normal form for an immersion

Theorem

Let $f : M^m \to N^n$ be an immersion, i.e. $\ker f_{*p} = \{0\}$ for all $p \in M$ ($m \leq n$). For any $p \in M$, $f(p) = q \in N$, there exist local coordinate systems (U, φ) about p and (V, ψ) about q s.t.

$$\psi \circ f \circ \varphi^{-1}(x_1, \ldots, x_m) = (x_1, \ldots, x_m, 0, \ldots, 0).$$

Locally, an immersion looks like the *standard embedding* of $\mathbb{R}^m$ into $\mathbb{R}^n$.

Proof of the normal form for an immersion (I)

Given arbitrary local coordinate systems $(\widetilde{U}, \widetilde{\varphi})$ and $(\widetilde{V}, \widetilde{\psi})$ about p and q, the local form for f_* is given by

$$f_* \left(\frac{\partial}{\partial x_i} \right) = \sum_{j=1}^{n} \left. \frac{\partial (y_j \circ f \circ \widetilde{\varphi}^{-1})}{\partial x_i} \right|_{\varphi(p)} \left. \frac{\partial}{\partial y_j} \right|_{f(p)}.$$

So the Jacobian of f_* is $D\widetilde{\psi} \circ f_* \widetilde{\varphi}^{-1}\big|_{\widetilde{\varphi}(p)}$. Write $F \triangleq \widetilde{\psi} \circ f \circ \widetilde{\varphi}^{-1}$. WLOG we can assume

$$\left(\frac{\partial F_i}{\partial x_j} \right)_{1 \le i \le m,\ 1 \le j \le m} \quad \text{is non-degenerate.}$$

Proof of the normal form for an immersion (II)

Define the map

$$\Phi(x_1, \ldots, x_m, x_{m+1}, \ldots, x_n) \triangleq \begin{pmatrix} F_1(x_1, \ldots, x_m) \\ \vdots \\ F_m(x_1, \ldots, x_m) \\ F_{m+1}(x_1, \ldots, x_m) + x_{m+1} \\ \vdots \\ F_n(x_1, \ldots, x_m) + x_n \end{pmatrix}.$$

Then

$$D\Phi = \begin{pmatrix} \left(\dfrac{\partial F_i}{\partial x_j}\right)_{m \times m} & 0 \\ * & I_{n-m} \end{pmatrix}$$

is non-degenerate.

Proof of the normal form for an immersion (III)

Therefore there are open subsets $U \subset \widetilde{\varphi}(\widetilde{U})$, $V \subset \widetilde{\psi}(\widetilde{V})$ such that $\Phi\big|_U : U \to V$ is a diffeomorphism, and

$$(x_1, \ldots, x_m, 0, \ldots, 0) \xleftrightarrow{\text{one-to-one}} \begin{pmatrix} F_1 \\ \vdots \\ F_n \end{pmatrix}.$$

If we consider the composition $\Phi^{-1} \circ F$, then it is clear that

$$\Phi^{-1} \circ F(x_1, \ldots, x_m) = (x_1, x_2, \ldots, x_m, 0, \ldots, 0).$$

Normal Form for a Submersion

The local normal form for a submersion

Theorem

Let $f : M^m \to N^n$ be a submersion, i.e. f_{*p} is surjective ($m \geq n$). Then for all $p \in M$ and $q = f(p) \in N$, there are local coordinate systems (U, φ) and (V, ψ) s.t. $\psi \circ f \circ \varphi^{-1} : \varphi(U) \to \psi(V)$ is given by

$$\psi \circ f \circ \varphi^{-1}(x_1, \ldots, x_m) = (x_1, \ldots, x_n).$$

Locally, a submersion looks like the *standard submersion* of $\mathbb{R}^m$ onto $\mathbb{R}^n$.

Proof of the normal form for a submersion (I)

For all $p \in M$ and $q = f(p) \in N$, take arbitrary local coordinate systems $(\widetilde{U}, \widetilde{\varphi})$ of p and $(\widetilde{V}, \widetilde{\psi})$ of q. Define

$$F = \widetilde{\psi} \circ f \circ \widetilde{\varphi}^{-1} : \widetilde{\varphi}(\widetilde{U}) \longrightarrow \widetilde{\psi}(\widetilde{V}).$$

Since f_* is a surjective map, $DF = \left(\frac{\partial F_i}{\partial x_j}\right)_{1 \le i \le n,\ 1 \le j \le m}$ has full rank (n).

Assume $\left(\frac{\partial F_i}{\partial x_j}\right)_{1 \le i \le n,\ 1 \le j \le n}$ is non-degenerate. Then define the map

$$\Phi(x_1; \ldots; x_m) = \begin{pmatrix} F_1 \\ \vdots \\ F_n \\ x_{n+1} \\ \vdots \\ x_m \end{pmatrix} = \begin{pmatrix} \widetilde{y}_1 \\ \vdots \\ \widetilde{y}_m \end{pmatrix}.$$

Proof of the normal form for a submersion (II)

Then

$$D\Phi = \begin{pmatrix} \left(\dfrac{\partial F_i}{\partial x_j}\right)_{n\times n} & * \\ 0 & I_{m-n} \end{pmatrix}$$

is non-degenerate. Therefore

$$\Phi^{-1}(\widetilde{y}_1,\dots,\widetilde{y}_m) = (x_1,\dots,x_m), \qquad \Phi\circ\Phi^{-1}(\widetilde{y}_1,\dots,\widetilde{y}_m) = (\widetilde{y}_1,\dots,\widetilde{y}_m),$$

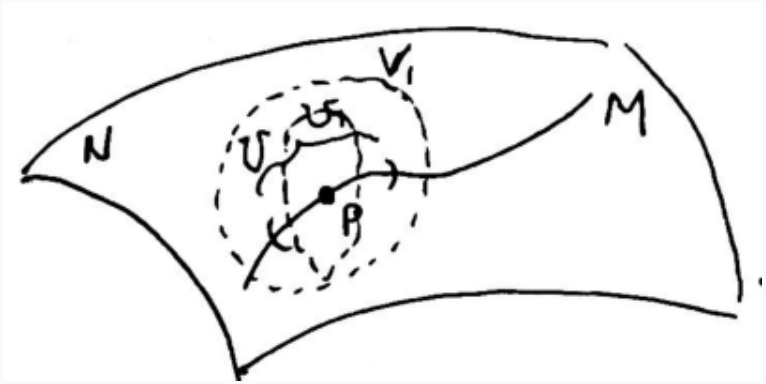
$$\Longrightarrow \quad F\circ\Phi^{-1}(\widetilde{y}_1,\dots,\widetilde{y}_m) = (\widetilde{y}_1,\dots,\widetilde{y}_n) = \big(F_1(x_1,\dots,x_m),\dots,F_n(x_1,\dots,x_m)\big).$$

So

$$\psi\circ f\circ\widetilde{\varphi}^{-1}\circ\Phi^{-1}(\widetilde{y}_1,\dots,\widetilde{y}_n) = (\widetilde{y}_1,\dots,\widetilde{y}_n).$$

Structure Theorem for Regular Submanifolds

The structure theorem



Theorem

M^m is a regular submanifold of N^n $\iff$ M is a subtopological space of N, and for all $p \in M$ there exists a local coordinate system (U, φ) of N with coordinate functions $(x_1, \ldots, x_n)$ s.t.

$$M \cap U = \{b \in U \mid x_i(b) = 0 \text{ for } m+1 \leq i \leq n\}.$$

i.e. locally M is cut out of N by the vanishing of the last $n - m$ coordinate functions.

Proof: embedding $\Rightarrow$ slice condition

Since $\iota : M \to N$ is an embedding, it is an immersion and a one-to-one map, and M is diffeomorphic to $\iota(M)$ with the subspace topology. By the local normal form for immersions, there are local coordinate systems (U_1, φ_1) of $p \in M$ and (V_1, ψ_1) of $p \in N$ s.t.

$$\psi_1 \circ \iota \circ \varphi_1^{-1}(x_1, \ldots, x_m) = (x_1, \ldots, x_m, 0, \ldots, 0).$$

Assume $\psi_1(q) = (x_1(q), \ldots, x_n(q))$, $q \in V_1$. Let

$$V_1' = \{q \in V_1 \mid (x_1(q), \ldots, x_m(q)) \in \varphi_1(U_1)\} \subset V_1.$$

Proof: shrinking the neighborhood

If we take the open set $U \subset V_1'$ of N such that $M \cap U \subset U_1$, then (V_1, ψ_1) restricts to (U, ψ_1) as a local coordinate system on N. For $q \in M \cap U \ (\subset U_1)$:

$$x_{m+1}(q) = \cdots = x_n(q) = 0, \qquad (x_1(q), \ldots, x_m(q)) \in \varphi_1(U_1).$$

Let $q' = \varphi_1^{-1}\big(x_1(q), \ldots, x_m(q)\big) \in U_1$. Then

$$\begin{aligned} \psi_1(q') &= \psi_1 \circ \iota \circ \varphi_1^{-1}\big(x_1(q), \ldots, x_m(q)\big) \\ &= \big(x_1(q), \ldots, x_m(q), 0, \ldots, 0\big) \\ &= \big(x_1(q), \ldots, x_m(q), x_{m+1}(q), \ldots, x_n(q)\big) = \psi_1(q). \end{aligned}$$

Since ψ_1 is one-to-one, $q = q' \in U_1 \subset M$. Hence $\{b \in U \mid x_{m+1}(b) = \cdots = x_n(b) = 0\} \subset M \cap U$.

Proof: slice condition $\Rightarrow$ embedding

If M satisfies: for all $p \in M$ there exists $(U; x_1, \ldots, x_n)$ of $p \in N$ s.t.

$$M \cap U = \{b \in U \mid x_{m+1}(b) = \cdots = x_n(b) = 0\},$$

then M is a local Euclidean space, with $M \cap U$ described by the coordinates $\{x_1(b), \ldots, x_m(b)\}$. Furthermore, M is a differentiable manifold, and the inclusion $\iota : M \to N$ is given by

$$(x_1, \ldots, x_m) \longmapsto (x_1, \ldots, x_m, 0, \ldots, 0),$$

which has full rank and is an embedding.

The Constant Rank Theorem

The constant rank theorem

Theorem (Constant Rank)

If $f : M^m \to N^n$ is a smooth map, and there exists a constant l s.t. $\operatorname{rank} f_{*p} = l$ *for all $p \in M$, then for all $q \in N$, $f^{-1}(\{q\})$ is either $\varnothing$ or a regular submanifold of M, with dimension $m - l$.*

Proof.

Let $S = f^{-1}(\{q\}) \subset M$; assume $S \neq \varnothing$. For all $p \in S \subset M$, let $q = f(p) \in N$. Then there are local coordinate systems (U, φ) and (V, ψ) s.t. $\varphi(p) = 0 \in \mathbb{R}^m$, $\psi(q) = 0 \in \mathbb{R}^n$, and

$$\psi \circ f \circ \varphi^{-1} : \varphi(U) \longrightarrow \psi(V)$$

$$(x_1, \ldots, x_m) \longmapsto \big(x_1, \ldots, x_l,\ g_{l+1}(x_1, \ldots, x_l), \ldots, g_n(x_1, \ldots, x_l)\big).$$

□

Proof of the constant rank theorem (I)

To prove this, consider $F : \mathbb{R}^m \to \mathbb{R}^n$ of rank l. Assume $\left(\frac{\partial F_i}{\partial x_j}\right)_{1\le i,j\le l}$ is non-degenerate. Define

$$\Phi(x_1,\dots,x_m) = \begin{pmatrix} F_1 \\ \vdots \\ F_l \\ x_{l+1} \\ \vdots \\ x_m \end{pmatrix} \implies D\Phi \text{ is non-degenerate.}$$

Since $\Phi \circ \Phi^{-1}(y_1,\dots,y_m) = (y_1,\dots,y_m)$,

$$F\circ\Phi^{-1}(y_1,\dots,y_m) = F(x_1,\dots,x_m) = \begin{pmatrix} F_1(x_1,\dots,x_m) \\ \vdots \\ F_n(x_1,\dots,x_m) \end{pmatrix} \triangleq \begin{pmatrix} y_1 \\ \vdots \\ y_l \\ g_{l+1}(x_1,\dots,x_m) \\ \vdots \end{pmatrix}.$$

Proof of the constant rank theorem (II)

Because $\operatorname{rank} D(F \circ \Phi^{-1}) = l = \operatorname{rank} DF$,

$$\frac{\partial g_i}{\partial y_j} = 0 \qquad \forall\, i = l+1, \ldots, n,\ \forall\, j = l+1, \ldots, m,$$

so $g_{l+1}(y_1, \ldots, y_l), \ldots, g_n(y_1, \ldots, y_l)$ depend only on $y_1, \ldots, y_l$.

Therefore there exists an open set $U \subset M$ s.t.

$$\begin{aligned} S \cap U &= \{s \in U \mid y_i(s) = 0 \text{ for } 0 < i \leq l\} \\ &= \{s \in U \mid y_1(s) = \cdots = y_l(s) = 0,\ g_{l+1}(y_1, \ldots, y_l) = 0, \ldots, g_n(y_1, \ldots, y_l) = 0\} \end{aligned}$$

By the structure theorem for regular submanifolds, S is a regular submanifold of M of dimension $m - l$.